Genome Variant Interpretation Report

Genome Variant Interpretation Assistant

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Executive Summary

This report presents an educational, ACMG-informed interpretation of the variant c.2573T>G identified in the EGFR gene. The variant has been characterized as a substitution change and is associated with a clinical significance of “drug response” based on the evidence sources consulted during this analysis, including ClinVar submissions, Ensembl functional annotation, and ACMG-style evidence weighting. Taken together, the available evidence supports a final interpretation of “Variant of Uncertain Significance”. As with any computationally assisted analysis, this classification is provisional and intended to illustrate the variant-interpretation workflow rather than to inform clinical decisions.

Variant Details

Field	Detail
Gene	EGFR
HGVS Notation	c.2573T>G
Variant Type	Substitution
Chromosome	7
Position	55191822
Transcript	ENST00000275493
Protein Change	ENSP00000275493.2:p.Leu858Arg
Genome Build	GRCh38

Gene Summary

This analysis focuses on EGFR. Ensembl reports the canonical transcript as ENST00000275493, located on chromosome 7. This section provides context for the variant interpretation and helps the reader understand where the change occurs in the gene.

Molecular Consequence

Field	Detail
Ensembl Consequence Term	missense_variant

Scientific Explanation

A missense variant changes a single codon so that it specifies a different amino acid than the reference sequence. The functional impact depends heavily on the chemical properties of the substituted residue and its location within the protein -- a change in a catalytic site or structural core is more likely to be disruptive than a change on a flexible, solvent-exposed surface.

Clinical Evidence

Field	Detail
ClinVar Significance	**drug response**
Review Status	reviewed by expert panel
Disease Association	gefitinib response - Efficacy

Why ClinVar Evidence Matters

ClinVar aggregates variant classifications submitted by clinical laboratories, research groups, and expert panels, along with the review status describing how thoroughly each submission was vetted. A higher review status (for example, assertions reviewed by an expert panel or reflecting a multi-submitter consensus) indicates greater confidence in the reported significance, while a single, unreviewed submission carries substantially less weight. Evaluating both the significance and the review status together -- rather than the significance alone -- is essential to correctly gauge how much confidence this evidence source should contribute to an overall interpretation.

ACMG Evidence Assessment

HGVS — Supporting (HGVS Validator)

A data-quality check confirming the variant description follows valid HGVS nomenclature; this affects confidence in downstream annotation rather than pathogenicity itself. Engine note: The variant description follows valid HGVS nomenclature, supporting confidence in the downstream annotation lookups.

5 pts

PVS1 — Not Applicable (Ensembl)

Very strong evidence of pathogenicity: the variant is predicted to cause loss of protein function (e.g. nonsense, frameshift, canonical splice-site, or initiation-codon change) in a gene where loss of function is an established disease mechanism. Engine note: Ensembl predicts missense_variant. Missense variants do not satisfy the PVS1 criterion.

0 pts

PM2 (gnomAD)

Moderate evidence of pathogenicity: the variant is absent, or present at only extremely low frequency, in population frequency databases such as gnomAD. Engine note: Population frequency data (gnomAD) were not available for this variant.

0 pts

PP3 (Computational Predictors)

Supporting evidence of pathogenicity: multiple computational (in-silico) prediction tools support a deleterious effect on the gene or protein product. Engine note: No computational prediction data (REVEL, AlphaMissense, SIFT, PolyPhen) were provided for this variant.

0 pts

ClinVar — Not Applicable (ClinVar)

'CLINVAR' does not follow a recognized ACMG/AMP evidence-code prefix. Refer to the originating evidence engine's documentation for its specific meaning. Engine note: ClinVar reports 'drug response', which is not directly used as pathogenic or benign evidence.

0 pts

FINAL CLASSIFICATION

Variant of Uncertain Significance

ACMG Evidence Score: 5 / 100 | Confidence: Moderate

Biological Interpretation

Based on the predicted missense_variant consequence, this variant may plausibly influence the structure, stability, or activity of the protein encoded by EGFR, though the direction and magnitude of any such effect cannot be determined from sequence-level annotation alone. Possible functional consequences range from negligible impact -- if the affected region tolerates substitution or truncation well -- to a meaningful disruption of protein function if the variant affects a critical domain, active site, or interaction surface. Whether such a disruption would meaningfully contribute to disease depends on additional factors not captured here, including the gene's mode of inheritance, tissue-specific expression, and the degree of functional redundancy with related genes. It is important to note that this assessment is inferential and does not constitute a claim of pathogenicity: confirming any real biological or clinical effect would require orthogonal functional studies, segregation data, or case-control evidence beyond the scope of this educational analysis.

Confidence Assessment

91 / 100

Component	Points	Rationale
HGVS Validation	25 / 25	The submitted HGVS notation was successfully validated.
ClinVar Review Quality	25 / 25	Reviewed by an expert panel or practice guideline -- the highest ClinVar review tier.
ACMG Evidence	16 / 25	The ACMG engine reported moderate confidence in its evidence weighting.
Ensembl Annotation Completeness	25 / 25	5 of 5 expected Ensembl annotation fields were populated.

How to Read This Score

This confidence score reflects the completeness and quality of the evidence gathered during analysis -- it is not a measure of how likely the variant is to be pathogenic. A low score most often indicates missing or low-quality annotation data rather than a benign variant, and a high score indicates that the interpretation rests on well-populated, higher-quality evidence sources.

Analysis Workflow

HGVS Validation

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Variant Classification

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Ensembl Annotation

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ClinVar Retrieval

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ACMG Evidence Evaluation

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Biological Interpretation

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Final Report

Educational Disclaimer

This software is intended for educational and research purposes only and is not suitable for clinical diagnosis or medical decision-making. The classifications, scores, and narrative interpretations in this report are generated by an automated, ACMG-inspired pipeline and have not been reviewed by a certified clinical laboratory, molecular pathologist, or genetic counselor. Any real-world clinical question regarding this or any other genetic variant should be directed to a qualified healthcare professional or accredited clinical genetics laboratory.